The independent domination number of the 23 by 23 toroidal queens graph: a computer-assisted proof and Lean formalization

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Abstract

We give a reproducible computer-assisted proof that the independent domination number of the 23×23 toroidal queens graph is 12. A displayed set of twelve queens gives the upper bound. For the lower bound, a finite-field coefficient-extraction argument first excludes dominating sets with at most ten queens. For eleven independent queens, translation to zero coordinate sums and seven coefficients of a degree-44 covering polynomial impose restrictions on the first three coefficients of four root polynomials. These restrictions reduce the remaining obstruction to candidate lists containing 2774 masks and a finite check of 40 620 104 triples. The entire implication from the chessboard definitions to the exact minimum is implemented in Lean 4 with mathlib. Three closed computations use `native_decide`; the compiler and native evaluation trust required by those steps is explicitly recorded. Two earlier finite enumerations and checked LRAT refutations provide additional evidence, but are not hypotheses of the Lean theorem. This manuscript makes no priority claim for the numerical value and concerns this fixed board size only.

Keywords: toroidal queens graph; independent domination; finite fields; polynomial method; exhaustive computation; Lean.

1 Problem, result, and scope

The vertices of Q_{23}^{t} are the pairs $(r, c) \in \mathbb{F}_{23}^2$. Distinct vertices (r, c) and (r', c') are adjacent when at least one of

$$r = r', \qquad c = c', \qquad r + c = r' + c', \qquad r - c = r' - c'$$

holds in $\mathbb{F}_{23}$. A queen covers its own square and its adjacent vertices. A set is *independent* if no two of its distinct queens attack each other; it is *dominating* if every square is covered. The independent domination number $i(Q_{23}^{\mathrm{t}})$ is the minimum cardinality of a set with both properties.

Theorem 1.1. *The independent domination number of the* 23×23 *toroidal queens graph is*

$$i(Q_{23}^{\mathrm{t}}) = 12.$$

Toroidal queens graphs and their domination parameters were studied by Burger, Mynhardt, and Cockayne [1], and upper bounds were developed by Mynhardt [2]. OEIS sequence

A279404 records the independent domination parameter [3]. The full text of [2] was checked: its main constructions concern orders divisible by 2 or 3, and Table 2 on page 166 lists independent domination values through order 16. It does not determine the value at order 23. The OEIS page obtained in this check lists values through order 21, with linked 2026 contributions for orders 19 and 20. A bounded literature search on November 8, 2026 did not locate an earlier proof of the exact value at order 23. This is not an exhaustive determination of priority, and no priority claim is made.

The proof has three parts. Section 3 excludes ten or fewer queens without assuming independence. Sections 4–5 exclude eleven independent queens using algebra followed by finite computation. Section 6 gives the twelve-queen construction. Section 7 describes the formal statements and their precise trust boundary. The result makes no assertion about an arbitrary board size n.

2 A coefficient-extraction lemma

All coefficients and evaluations in this section lie in $\mathbb{F}_{23}$. For $n \geq 0$, put $\sigma_n = \sum_{t\in\mathbb{F}_{23}} t^n$, with the usual polynomial convention $t^0 = 1$. Then

$$\sigma_n = \begin{cases} -1, & n > 0 \text{ and } 22 \mid n, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

For $n = 0$ this is $23 = 0$ in $\mathbb{F}_{23}$; for positive n, it follows by summing powers in the cyclic multiplicative group of order 22.

Lemma 2.1. *Let $P \in \mathbb{F}_{23}[x, y]$ have total degree at most 44, and suppose that $P(x, y) = 0$ for every $(x, y) \in \mathbb{F}_{23}^2$. If $0 \leq a, b \leq 22$ and $a + b > 22$, then $[x^a y^b]P = 0$.*

Proof. Consider the linear functional

$$\Lambda_{a,b}(P) = \sum_{x\in\mathbb{F}_{23}} \sum_{y\in\mathbb{F}_{23}} x^{22-a} y^{22-b} P(x, y).$$

On a monomial $x^i y^j$ its value is $\sigma_{22-a+i}\sigma_{22-b+j}$. If this value is nonzero, both subscripts are positive multiples of 22. Thus $i = a + 22u$ and $j = b + 22v$ for integers $u, v \geq 0$. Since $i + j \leq 44$ and $a + b > 22$, we have $u = v = 0$. For $(i, j) = (a, b)$ the value is $(-1)^2 = 1$. Consequently $\Lambda_{a,b}(P) = [x^a y^b]P$. The assumed vanishing of P makes the left side zero. □

For a finite set $Z \subseteq \mathbb{F}_{23}$, define its root polynomial by

$$L_Z(T) = \prod_{z\in Z} (T - z).$$

For four sets $R, C, S, D \subseteq \mathbb{F}_{23}$, define the covering polynomial

$$P_{R,C,S,D}(x, y) = L_R(x)L_C(y)L_S(x + y)L_D(x - y). \tag{2}$$

If R, C, S, D contain, respectively, the row, column, sum-diagonal, and difference-diagonal labels of a dominating set of queens, then this polynomial vanishes on $\mathbb{F}_{23}^2$: a covering queen makes at least one of its four factors zero.

3 Excluding ten queens or fewer

Proposition 3.1. *Every dominating set in Q_{23}^{t}, whether independent or not, has at least eleven queens.*

Proof. Suppose that a dominating set has at most ten queens. Each of its four projection sets has size at most ten. Enlarge each projection independently to a subset R, C, S, D of $\mathbb{F}_{23}$ of size exactly ten; this is possible since $|\mathbb{F}_{23}| = 23$. The polynomial in (2) still vanishes on $\mathbb{F}_{23}^2$. It has total degree at most 40, and its homogeneous part of degree 40 is

$$x^{10}y^{10}(x+y)^{10}(x-y)^{10} = x^{10}y^{10}(x^2-y^2)^{10}.$$

All other terms have degree at most 39. Hence

$$[x^{20}y^{20}]P_{R,C,S,D} = (-1)^5\binom{10}{5} = -252 = 1 \quad \text{in } \mathbb{F}_{23}.$$

Lemma 2.1, with $(a, b) = (20, 20)$, says that the same coefficient must be zero. This contradiction proves the proposition. □

4 Necessary conditions for eleven independent queens

Suppose for contradiction that $A \subseteq \mathbb{F}_{23}^2$ is independent and dominating and $|A| = 11$. Write its elements as (r_i, c_i), and put

$$R = \{r_i\}, \quad C = \{c_i\}, \quad S = \{r_i + c_i\}, \quad D = \{r_i - c_i\}.$$

Independence makes each of the four projection maps injective on A; therefore every one of these sets has size 11.

4.1 Centering and root coefficients

Translate all queens by

$$\left(-11^{-1}\sum_i r_i, -11^{-1}\sum_i c_i\right).$$

Here $11^{-1} = 21$ in $\mathbb{F}_{23}$. Translation preserves cardinality, independence, and domination. We may therefore assume

$$\sum_{r\in R} r = \sum_{c\in C} c = \sum_{s\in S} s = \sum_{d\in D} d = 0. \tag{3}$$

No queen is required to be at the origin after this translation.

For each of the four sets Z, write

$$L_Z(T) = T^{11} + u_Z T^9 + v_Z T^8 + E_Z(T), \qquad \deg E_Z \le 7. \tag{4}$$

The missing T^{10} term follows from (3). The coefficients can equivalently be read from

$$B_Z(T) = \prod_{z\in Z}(1 - zT): \qquad u_Z = [T^2]B_Z, \quad v_Z = [T^3]B_Z. \tag{5}$$

The elementary coefficient identity

$$2[T^2]B_Z = \left(\sum_{z\in Z} z\right)^2 - \sum_{z\in Z} z^2 \tag{6}$$

follows either by expanding the product or by induction when one root is inserted. Under (3), it gives $2u_Z = -\sum_{z\in Z} z^2$.

4.2 Seven coefficients of the covering polynomial

The polynomial $P = P_{R,C,S,D}$ has degree at most 44 and vanishes on the whole board. By Lemma 2.1, its coefficients at

$$(22,20),(21,21),(20,22), \quad (22,19),(21,20),(20,21),(19,22) \tag{7}$$

are all zero.

For $j \in \{2,3\}$, define the four homogeneous kernels

$$\begin{aligned} K_{R,j} &= x^{11-j}y^{11}(x+y)^{11}(x-y)^{11}, \\ K_{C,j} &= x^{11}y^{11-j}(x+y)^{11}(x-y)^{11}, \\ K_{S,j} &= x^{11}y^{11}(x+y)^{11-j}(x-y)^{11}, \\ K_{D,j} &= x^{11}y^{11}(x+y)^{11}(x-y)^{11-j}. \end{aligned}$$

The degree-42 part of P is $\sum_Z u_Z K_{Z,2}$ and its degree-41 part is $\sum_Z v_Z K_{Z,3}$. Indeed, a single u_Z term drops the degree by two, a single v_Z term drops it by three, a tail term drops it by at least four, and two or more nonleading factors also drop it by at least four. Thus no cross term has been omitted in degrees 42 and 41.

Straight binomial expansion gives the following coefficient matrices over $\mathbb{F}_{23}$. The column order in both cases is R, C, S, D.

	R	C	S	D
$x^{22}y^{20}$	0	0	1	22
$x^{21}y^{21}$	21	2	0	0
$x^{20}y^{22}$	0	0	22	1

(8)

	R	C	S	D
$x^{22}y^{19}$	8	0	17	17
$x^{21}y^{20}$	0	2	7	16
$x^{20}y^{21}$	21	0	16	16
$x^{19}y^{22}$	0	15	6	17

(9)

These are exact polynomial calculations, not tests on sampled queen sets. The accompanying Lean files prove characteristic-free sparse expansions using `ring`, then extract and reduce the specified coefficients.

The first two equations in (8) give

$$u_R = u_C, \qquad u_S = u_D. \tag{10}$$

The matrix in (9) is invertible over $\mathbb{F}_{23}$. For example an inverse is

$$\begin{pmatrix} 16 & 0 & 6 & 0 \\ 0 & 17 & 0 & 7 \\ 1 & 19 & 4 & 22 \\ 1 & 4 & 4 & 1 \end{pmatrix}. \tag{11}$$

Consequently

$$v_R = v_C = v_S = v_D = 0. \tag{12}$$

There is one further geometric identity. Injectivity of the projections and $(r+c)^2 + (r-c)^2 = 2(r^2+c^2)$ imply

$$\sum_{s\in S} s^2 + \sum_{d\in D} d^2 = 2\left(\sum_{r\in R} r^2 + \sum_{c\in C} c^2\right).$$

Using (6) and (10), and cancelling the nonzero element 2, gives

$$u_S = u_D = 2u_R = 2u_C. \tag{13}$$

In particular, all the conditions used below are necessary for an eleven-queen independent dominating set. We make no symmetry assumption about that set.

5 The finite obstruction

For $b \in \mathbb{F}_{23}$, let $\mathcal{P}_b$ be the family of sets $Z \subseteq \mathbb{F}_{23}$ satisfying

$$|Z| = 11, \quad \sum_{z \in Z} z = 0, \quad [T^2]B_Z = b, \quad [T^3]B_Z = 0. \tag{14}$$

The preceding section gives $R, C \in \mathcal{P}_b$ and $S, D \in \mathcal{P}_{2b}$ for $b = u_R$.

For any three sets of line labels, put

$$\Delta(R, C, S) = \{d \in \mathbb{F}_{23} : \exists r \in \mathbb{F}_{23}, r \notin R,\ r - d \notin C,\ 2r - d \notin S\}. \tag{15}$$

Equivalently these are the difference labels of squares left uncovered by the row, column, and sum-diagonal families. If the four line families cover the board, then

$$\Delta(R, C, S) \subseteq D. \tag{16}$$

Indeed, for a witness r in (15), the square $(r, r - d)$ has no covering row, column, or sum diagonal, so must be covered by difference label d.

Lemma 5.1 (finite computation)**.** *For all $b \in \mathbb{F}_{23}$ and all $R, C \in \mathcal{P}_b$, $S \in \mathcal{P}_{2b}$,*

$$|\Delta(R, C, S)| \geq 12.$$

Here the finite computation is part of the proof. Its definition, exhaustive scope, and connection to the stated sets are described next.

5.1 Candidate lists and certified completeness

Encode $Z \subseteq \mathbb{F}_{23}$ as the natural number $m(Z) = \sum_{z \in Z} 2^{\mathrm{val}(z)}$, where residues are represented by $0, \ldots, 22$. The file `ProfileData.lean` contains 23 explicit candidate lists $\mathcal{L}_b$. To prove that every profile is in the appropriate list, an executable Lean function scans *every* mask $0 \leq m < 2^{23} = 8\,388\,608$.

For a partial product $\prod(1 - zT)$, keep a state (k, a, b, c) consisting of the number of roots and coefficients of T, T^2, T^3. Inserting a root z updates this state by

$$(k, a, b, c) \longmapsto (k + 1, a - z, b - za, c - zb). \tag{17}$$

The initial state is $(0, 0, 0, 0)$, and a root is inserted precisely when the corresponding bit of m is set. For every mask with $k = 11$, $a = 0$, and $c = 0$, the checker requires $m \in \mathcal{L}_b$. The scan is evaluated by `native_decide`. An ordinary Lean induction proves that this recurrence equals the coefficients of (5), and another proves that a successful range scan checks each integer in its range. Together these establish

$$Z \in \mathcal{P}_b \implies m(Z) \in \mathcal{L}_b. \tag{18}$$

The proof needs this inclusion only; it does not assume that external list generation is correct or require the converse inclusion as a hypothesis.

There are 2774 entries in the lists. Their sizes and the resulting triple counts are shown in Table 1. The coefficient parameter b is not the second power sum: when the sum is zero, $b = -\frac{1}{2}\sum z^2 = 11\sum z^2$ in $\mathbb{F}_{23}$.

Coefficient parameter b	Number of groups	Entries per group	Triples in all groups
0	1	134	2 406 104
Nonzero quadratic residue	11	115	16 729 625
Quadratic nonresidue	11	125	21 484 375
Total	23	2774 entries	40 620 104

Table 1: Candidate list sizes and exhaustive coverage checks. Since 2 is a square modulo 23, b and $2b$ have the same group size.

5.2 The executable coverage check and its semantics

Masks in this computation are 23-bit vectors. If M is a mask, $\mathrm{rot}_t(M)$ denotes cyclic left rotation by t places; it represents the set obtained by adding t modulo 23. Let $H(S)$ be the mask whose bit at r is the bit of S at $2r$. It represents $S/2$, since $2^{-1} = 12$ in $\mathbb{F}_{23}$. For each $d \in \{0, \ldots, 22\}$, form the mask

$$U_d = \overline{R} \;\&\; \overline{\mathrm{rot}_d(C)} \;\&\; \overline{\mathrm{rot}_{12d}(H(S))}. \tag{19}$$

Complements are taken within 23 bits. Its bit at r is set exactly when $r \notin R$, $r - d \notin C$, and $2r - d \notin S$. Therefore $d \in \Delta(R, C, S)$ if and only if $U_d \neq 0$.

The executable checker loops over all 23 values of b, every column mask in $\mathcal{L}_b$, every sum mask in $\mathcal{L}_{2b}$, and every row mask in $\mathcal{L}_b$. For each triple it counts the 23 indices with $U_d \neq 0$ and requires the count to be at least 12. A closed `native_decide` theorem certifies that every check returns true. Rotations and allowed-row masks are cached outside the inner row loop. All groups are checked; the Lean proof uses no scaling reduction.

The file `CoverSemantics.lean` proves that this executable count is exactly the cardinality of (15). In particular, the proof relates bit-vector rotations, residue subtraction, set encoding, and finite-set cardinality. Applying this bridge to the successful exhaustive check and using (18) proves Lemma 5.1.

5.3 The contradiction

For the hypothesized eleven-queen set, $|D| = 11$ and (16) give $|\Delta(R, C, S)| \leq 11$. Lemma 5.1 gives the opposite bound $|\Delta(R, C, S)| \geq 12$. This excludes eleven independent dominating queens. With Proposition 3.1, every independent dominating set has at least twelve queens.

6 A twelve-queen construction

All coordinates below are zero indexed and interpreted modulo 23. Take the twelve squares

$$\begin{aligned}&(2, 13), (3, 9), (6, 3), (7, 22), (9, 2), (11, 5),\\ &(12, 8), (15, 18), (16, 14), (18, 7), (20, 4), (21, 0).\end{aligned} \tag{20}$$

There are twelve distinct rows, columns, sum labels, and difference labels, so the queens are independent. Direct evaluation on the 529 squares shows that they dominate the board. Both assertions, and cardinality twelve, are proved by ordinary `decide` in `Upper.lean`; this upper-bound proof does not use `native_decide`. Combined with the lower bound, it proves Theorem 1.1.

7 Lean implementation and trust boundary

7.1 Formal statements and dependency structure

The implementation uses Lean 4.34.0 [4] and mathlib [5] source revision

`5ed2965256430c3649e86755f9576b54eca72435`.

The proof models the field as `ZMod 23` and squares as pairs of field elements. Its definitions of independence and domination quantify over the actual squares and queens, rather than assuming any polynomial equations or profile restrictions. The main exact-minimum statement is

```
theorem minimum_is_twelve :
    (exists s : Finset Square,
      s.card = 12 /\ Independent s /\ Dominating s) /\
    (forall s : Finset Square,
      Independent s -> Dominating s -> 12 <= s.card)
```

Here the ASCII display gives the logical content of the Unicode source. The source also defines the minimum by the infimum of attainable cardinalities and proves the equality

```
theorem independentDominationNumber_eq_twelve :
    independentDominationNumber = 12
```

Both the existence and universal lower bound are unconditional within the declared Lean/-native computation trust boundary.

The modules have the following roles.

Module(s)	Content
`Basic`	Chessboard definitions, translations, centering, projection injectivity.
`RootCoefficients`	Root products, reversal, coefficient identities and tails.
`Grid`, `Lift`	Grid coefficient extraction, substitution and degree bounds.
`Kernels`, `Expansion`, `Equations`	Exact expansions, error-degree bounds, seven coefficients and linear elimination.
`Geometry`, `LowerTen`	Projection identities, coefficient restrictions, and exclusion of at most ten queens.
`ProfileData`, `ProfileEnumeration`, `Masks`, `ProfileBridge`	Explicit lists, exhaustive completeness, mask semantics and coefficient recurrence.
`CoverCheck`, `CoverSemantics`	Exhaustive triple check and its interpretation as $\lvert\Delta\rvert$.
`Upper`, `Main`	Twelve-queen witness and the complete minimum theorem.

Table 2: Structure of the accompanying Lean source. Module names are relative to the namespace `ToroidalQueens23`.

7.2 Native computation is explicitly trusted

The final build passed with no `sorry`, `admit`, or manually declared axiom in the proof source. The algebraic identities, geometric bridges, recurrences, and range-scan soundness are ordinary Lean proofs. Exactly three closed calculations use `native_decide`:

1. `powerSum_table`: the power sums for $0 \le n < 89$;
2. `all_profile_masks_checked`: completeness over all 2^{23} masks;

3. `all_profile_covers_checked`: all 23 candidate groups and their 40 620 104 triples.

The axiom report for the final equality includes `propext`, `Classical.choice`, and `Quot.sound`, together with the three generated native computation axioms named in Appendix A. There is no `sorryAx` in that report.

This is a complete Lean-compiled formalization with a native computation trust component. It is not claimed to be a proof reduced entirely by the Lean kernel using only the three standard axioms. Correctness of the three native steps additionally relies on the relevant compiler/runtime and native evaluation mechanism. The finite computations are closed Lean programs with proved semantic bridges; no Python result or SAT solver status is imported as an assumed theorem.

7.3 Supplementary checks and reproducibility

Before the complete formalization, two different implementations checked the finite obstruction independently. One enumerated all $\binom{23}{11} = 1\,352\,078$ eleven-element subsets and accumulated uncovered difference labels directly. The other used fixed-weight integer masks and vectorized integer rotations. Both checked all 40 620 104 triples; the second reported a minimum of 13 required difference labels, and 15 in the zero-parameter group. These stronger observed minima are supplementary results, not the bound certified or needed by the Lean theorem.

Three additional profile SAT instances, for second-moment representatives $0, 1, 22$, yielded LRAT refutations. A checker separate from the solver verified all three to the empty clause, totaling 1 867 319 clause additions and 139 363 759 positive hint references. These historical certificates and their logs are retained. The Lean proof instead checks every coefficient group directly and does not formalize or depend on those LRAT certificates. A known five-queen construction on the 11×11 torus was also used as a positive control for the earlier programs.

The source distribution includes the proof modules, exact candidate data, toolchain pin, a portable Lake configuration, build/axiom audit, and checksums. The manuscript source uses standard LaTeX packages. Rebuilding the formal proof with a fresh mathlib environment requires installing the pinned Lean toolchain, fetching dependencies, and running `lake build`. A successful build re-evaluates the closed native checks when their modules are rebuilt; on the Windows development environment, the coverage computation took about 12 minutes. Timings depend on hardware and are not proof hypotheses. The checksums identify the exact tested source, and the build log distinguishes newly compiled modules from cached ones. Historical failed development logs are preserved separately from the final successful audit.

8 Limitations and further work

The result is a fixed finite instance of independent domination on a torus. The covering-polynomial restrictions are necessary conditions; the proof does not classify all optimal configurations, prove a general formula in n, or establish a geometric symmetry of hypothetical critical configurations. The current finite obstruction is computational. Replacing it by a short structural proof, or replacing native evaluation by compact kernel-checkable certificates, would strengthen the proof's explanatory or trust properties.

This manuscript is a preprint draft. The numerical value's literature priority has not been established, and no external subject-expert review is claimed. The proof and files were prepared with assistance from OpenAI Codex for mathematical development, programming,

formalization, and writing. Such assistance or agreement is not evidence in place of the proofs and executable checks specified above.

A Exact native axiom audit

With the common prefix `ToroidalQueens23.` omitted, the generated native names appearing in the final equality's axiom report are

```
all_profile_covers_checked._native.native_decide.ax_1_1
all_profile_masks_checked._native.native_decide.ax_1_1
powerSum_table._native.native_decide.ax_1_1
```

The remaining names are `propext`, `Classical.choice`, and `Quot.sound`. The upper bound alone has only those three standard names. The source bundle includes the literal report rather than treating this prose as its proof certificate.

B Coefficient formula for reproducing the kernel tables

For nonnegative integers a, b, c, d, the coefficient of $x^i y^j$ in $x^a y^b (x+y)^c (x-y)^d$ is

$$\sum_{\substack{0\le k\le c,\ 0\le \ell\le d \\ a+c+d-k-\ell=i,\ b+k+\ell=j}} \binom{c}{k}\binom{d}{\ell}(-1)^{\ell}.$$

Apply this formula with the exponent tuples

$$(11-j,11,11,11),\ (11,11-j,11,11),\ (11,11,11-j,11),\ (11,11,11,11-j),$$

for $j = 2, 3$, and reduce the resulting integers modulo 23. The Lean source proves the full integer-coefficient expansions by ring normalization before extracting the table entries, so the generator that emits these expressions is not a trusted oracle.

References

[1] A. P. Burger, C. M. Mynhardt, and E. J. Cockayne, *Queens graphs for chessboards on the torus*, Australasian Journal of Combinatorics **24** (2001), 231–246. https://ajc.maths.uq.edu.au/pdf/24/ocr-ajc-v24-p231.pdf.

[2] C. M. Mynhardt, *Upper bounds for the domination numbers of toroidal queens graphs*, Discussiones Mathematicae Graph Theory **23** (2003), 163–175. doi:10.7151/dmgt.1193.

[3] The OEIS Foundation Inc., *A279404: Independent domination number for queens' graph on an $n \times n$ toroidal board*, accessed November 8, 2026. https://oeis.org/A279404.

[4] The Lean development team, *Lean 4*, version 4.34.0. https://github.com/leanprover/lean4.

[5] The mathlib community, *mathlib4*, source revision `5ed2965256430c3649e86755f9576b54eca72435`. https://github.com/leanprover-community/mathlib4.